

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# STRAIGHT-LINE GRAPHS, (EQUATION OF A STRAIGHT LINE)

LO: I can find and use the equation of a straight line in the form  $y = mx + c$ .

$$y = mx + c$$

The equation of a straight line is  $y = mx + c$ , where **m** is the gradient (slope) and **c** is the y-intercept (where the line crosses the y-axis).

**Gradient:**  $m = \text{rise} \div \text{run} = \text{change in } y \div \text{change in } x$  between any two points on the line.

## Quick examples

|   |                                           |                      |
|---|-------------------------------------------|----------------------|
| ● | $y = 2x + 3$ : gradient 2, y-intercept 3  | read m and c         |
| ● | $y = -x + 5$ : gradient -1, y-intercept 5 | negative gradient    |
| ● | $m = (y_2 - y_1)/(x_2 - x_1)$             | gradient formula     |
| ● | $y = 2x + 3$ has gradient 3               | gradient is 2, not 3 |
| ● | $y = 4 - 2x$ has negative gradient        | gradient is 2        |

## Key idea

Every straight line can be written as  $y = mx + c$ . Find m from two points, then substitute a point to find c.

$$m = \frac{(y_{\text{sub}2} - y_{\text{sub}1})}{(x_{\text{sub}2} - x_{\text{sub}1})} \div$$

Then use  $y = mx + c$  with a known point to find c

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - reading from equation

State the gradient and y-intercept of  $y = 3x - 2$ .

Compare with  $y = mx + c$

$m = 3$  (gradient)

$c = -2$  (y-intercept)

**Gradient = 3, y-intercept = -2**

The coefficient of x is m; the constant is c.

### Example 2 - finding equation from graph

A line passes through (0, 4) and (2, 10). Find its equation.

$c = 4$  (y-intercept)

$m = (10 - 4)/(2 - 0) = 6/2 = 3$

Equation:  $y = 3x + 4$

**$y = 3x + 4$**

Use the y-intercept for c and calculate m from two points.

### Example 3 - two points

Find the equation of the line through (1, 5) and (3, 11).

$m = (11 - 5)/(3 - 1) = 6/2 = 3$

Using (1, 5):  $5 = 3(1) + c$ ,  $c = 2$

Equation:  $y = 3x + 2$

**$y = 3x + 2$**

Find gradient first, then substitute a point to find c.

### Example 4 - rearranging

Write  $2x + 3y = 12$  in the form  $y = mx + c$ .

$3y = 12 - 2x$

$3y = -2x + 12$

$y = -2/3x + 4$

**$y = -2/3x + 4$**

Rearrange to isolate y on the left-hand side.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Reading m and c

State the gradient and y-intercept of each line.

- a)  $y = 5x + 1$
- b)  $y = -2x + 7$
- c)  $y = \frac{1}{2}x - 3$
- d)  $y = x$

### 2. Finding gradient from two points

Calculate the gradient of the line through each pair of points.

- a) (1, 3) and (3, 7)
- b) (0, 5) and (4, 1)
- c) (2, 1) and (6, 9)
- d) (-1, 4) and (3, 0)

### 3. Finding the equation

Find the equation  $y = mx + c$  for each line.

- a) Through (0, 3) with gradient 2
- b) Through (0, -1) with gradient 4
- c) Through (1, 5) and (3, 11)
- d) Through (2, 3) and (4, 9)

### 4. Rearranging equations

Write each equation in the form  $y = mx + c$ .

- a)  $2y = 6x + 4$
- b)  $x + y = 8$
- c)  $3x + 2y = 10$
- d)  $4x - y = 3$

### 5. Mixed problems

Solve each problem.

- a) Does (2, 7) lie on  $y = 3x + 1$ ?
- b) Find where  $y = 2x + 1$  crosses the x-axis.
- c) Line through (-1, 2) and (2, 8). Equation?
- d) Line  $y = 4x + c$  passes through (3, 10). Find c.
- e) Where do  $y = x + 3$  and  $y = 2x + 1$  intersect?

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) In  $y = mx + c$ , m is the gradient and c is the y-intercept ☐

Correct answer: \_\_\_\_\_  
Reason: \_\_\_\_\_

b)  $y = 3 - 2x$  has gradient 3 ☐

Correct answer: \_\_\_\_\_  
Reason: \_\_\_\_\_

c) The gradient is the change in y divided by the change in x ☐

Correct answer: \_\_\_\_\_  
Reason: \_\_\_\_\_

d)  $y = 2x + 5$  crosses the y-axis at (5, 0) ☐

Correct answer: \_\_\_\_\_  
Reason: \_\_\_\_\_

e) A line with gradient 0 is horizontal ☐

Correct answer: \_\_\_\_\_  
Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A line passes through the points (-2, -3) and (4, 9). Find its equation. Determine whether the point (6, 13) lies on this line.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Two lines  $y = 2x + 1$  and  $y = -x + 7$  intersect at point P. Find the coordinates of P and the equation of the line through P that is parallel to  $y = 3x - 4$ .

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

